Cambridge (CIE) A Level Maths: Pure 3

Your notes

Modulus Functions

Contents

- Sketching Graphs of Modulus Functions
- Solving Equations with Modulus Functions

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Your notes

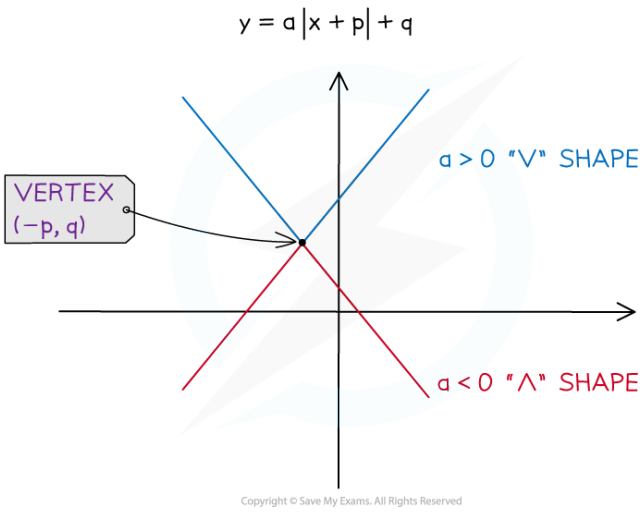
Sketching graphs of modulus functions

What is the modulus function?

- The **modulus** function makes **any** 'input' **positive**
- **$|x| = x$ if $x \geq 0$ $|f(x)| = f(x)$ if $f(x) \geq 0$**
- **$|x| = -x$ if $x < 0$ $|f(x)| = -f(x)$ if $f(x) < 0$**
  - **For example:** $|5| = 5$ and $|-5| = 5$
- Sometimes called **absolute value**

How do I sketch the straight line modulus graph $y = a|-x+p|+q$?

- The graph will look like a "≡" if $a > 0$ or a "≡" if $a < 0$
- There will be a vertex at the point $(-p, q)$
- There could be 0, 1 or 2 roots
  - This depends on the location of the vertex and the orientation of the graph (≡ or ≡)
- Compare this to the **completed square form** of a **quadratic** $a(x+p)^2 + q$

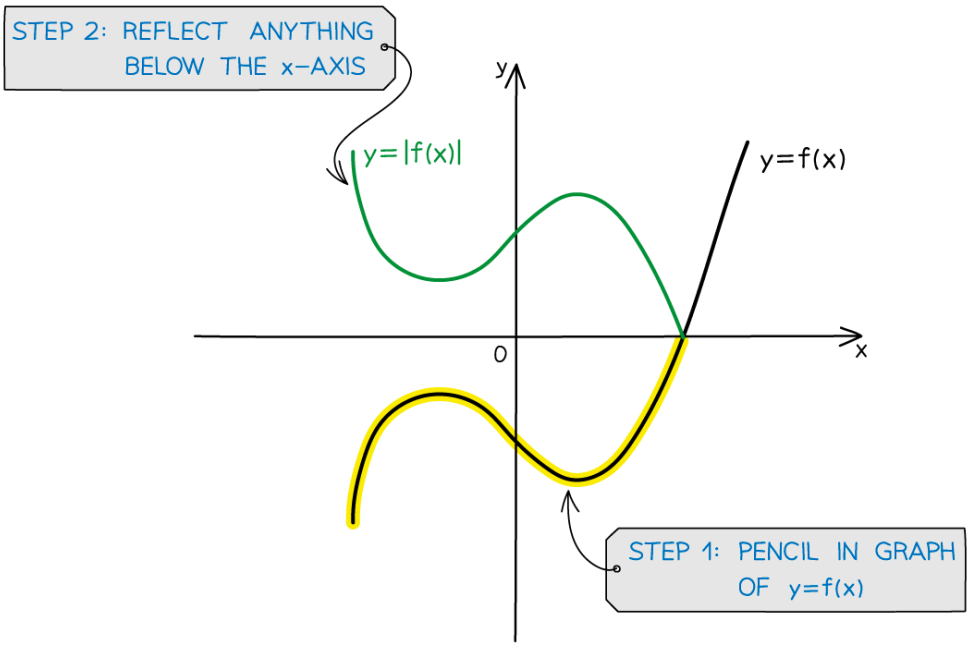


How do I sketch the modulus of a function $y = |f(x)|$?

STEP 1 Pencil in the graph of **$y = f(x)$**

STEP 2 **Reflect** anything **below** the **x**-axis, in the **x**-axis, to get **$y = |f(x)|$**

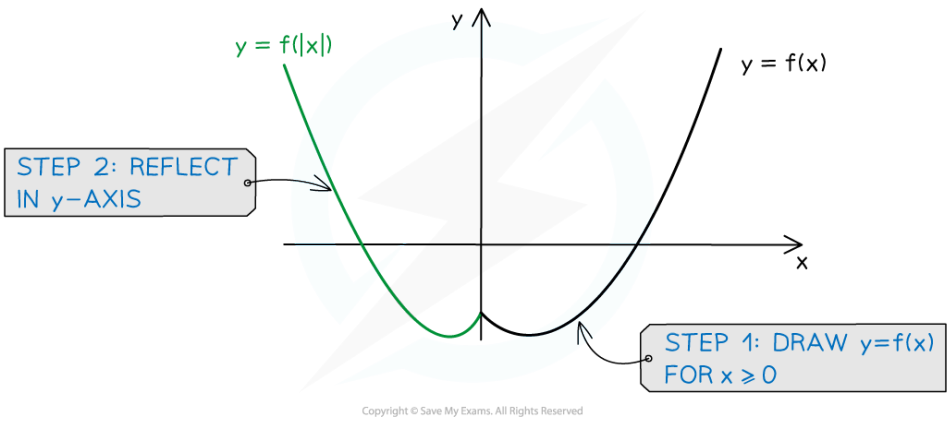
Your notes



How do I sketch a function of mod x, y = f(|x|)?

STEP 1 Sketch the graph of **y = f(x) only for $x \geq 0$**

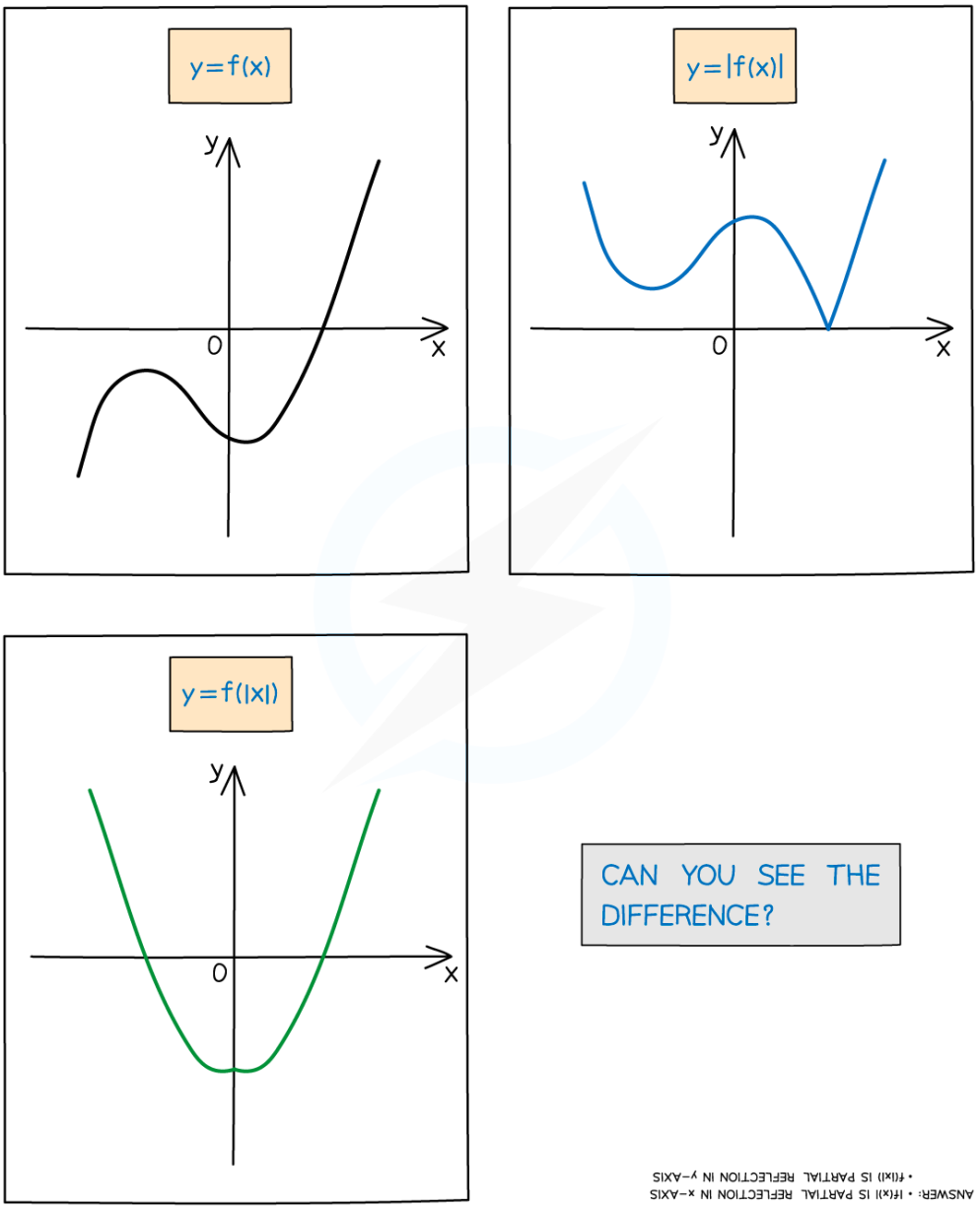
STEP 2 **Reflect** this in the **y-axis**



What is the difference between y = |f(x)| and y = f(|x|)?

- There is a difference between **y = |f(x)|** and **y = f(|x|)**
- The graph of **y = |f(x)| never** goes **below** the **x-axis**
  - It does not have to have any lines of symmetry

- The graph of $y = f(|x|)$ is **always symmetrical** about the y-axis
  - It can go below the x-axis



ANSWER: • |f(x)| IS PARTIAL REFLECTION IN x-AXIS
• f(|x|) IS PARTIAL REFLECTION IN y-AXIS

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- For the exam you will only be asked to do this when f(x) is **linear**
  - Your graphs will all look like a "≡" or a "≡"
  - You can also think of these graphs as **transformations** of the graph $y = |x|$

Your notes

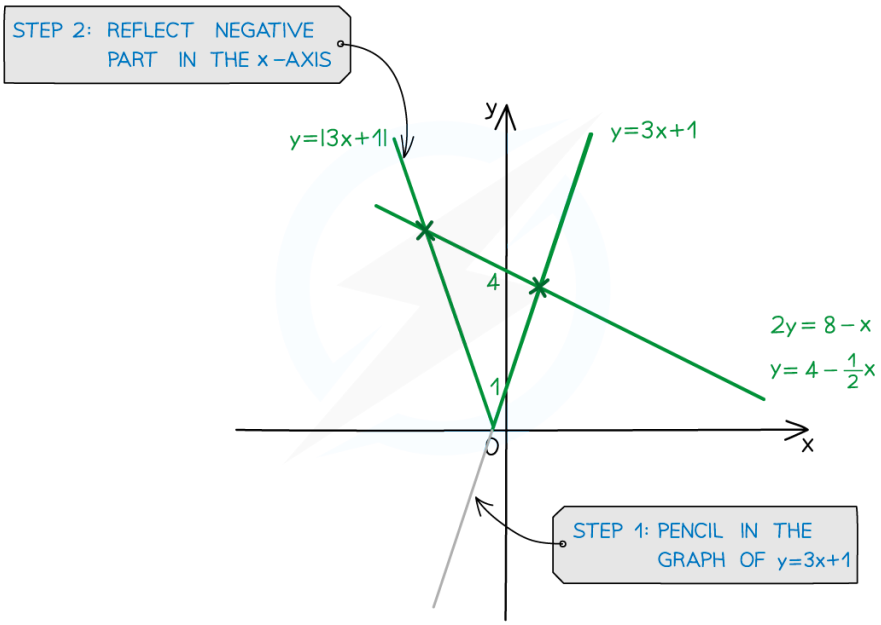
Worked Example

Your notes

(a) On the same diagram sketch the graphs of $y = |3x + 1|$ and $2y = 8 - x$

(b) Hence write down how many solutions there are to the equation $4 - \frac{1}{2}x = |3x + 1|$

(a)



(b) 2

THE GRAPHS CROSS TWICE

$4 - \frac{1}{2}x = -(3x+1)$

$4 - \frac{1}{2}x = 3x+1$

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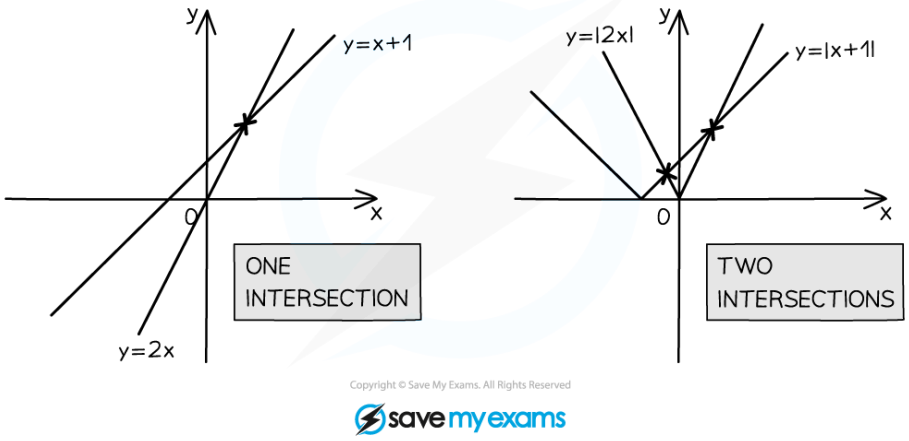
Solving equations with modulus functions

How do I sketch modulus equations?

- Two non-parallel straight-line graphs would intersect **once**
- If a **modulus** is involved there **could** be **more** than one intersection
- **Deducing** where these **intersections** are is crucial to solving equations



How do I solve modulus equations?

- STEP 1 **Sketch** the graphs including any **modulus** (reflected) parts
  - (see Modulus Functions – Sketching Graphs)
- STEP 2 **Locate** the graph **intersections**
- STEP 3 **Solve** the appropriate equation(s) or inequality
  - For $|f(x)| = |g(x)|$ the two possible equations are $f(x) = g(x)$ and $f(x) = -g(x)$

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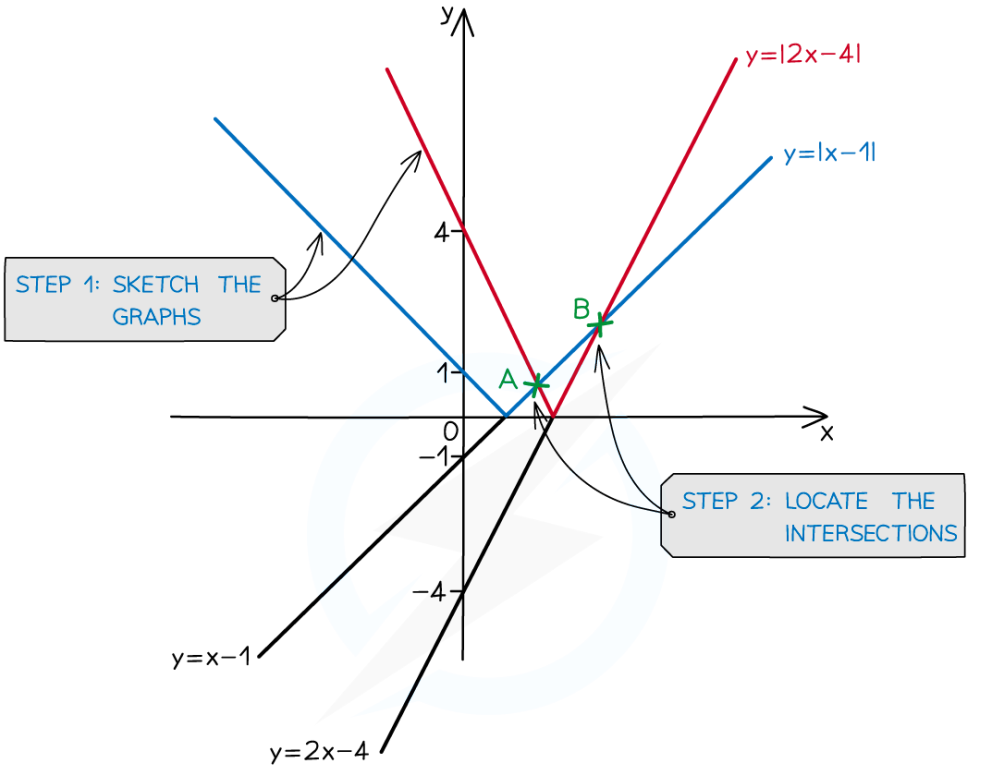
e.g. SOLVE $|2x-4| = |x-1|$



STEP 3: SOLVE THE APPROPRIATE EQUATIONS

A: $x-1=-(2x-4)$
$3x=5$
$x=\frac{5}{3}$

- "NORMAL" PART OF $x-1$
- "REFLECTED" PART OF $2x-4$

B: $x-1=2x-4$
$x=3$

"NORMAL" PART FOR BOTH

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Examiner Tips and Tricks

- Sketching the graphs is important as solving algebraically can lead to invalid solutions.
- For example, $x = 1$ is a solution to $x-4=2x-5$ but it is not a solution to $|x-4|=2x-5$ (substitute $x = 1$ into both sides and see why it does not work).

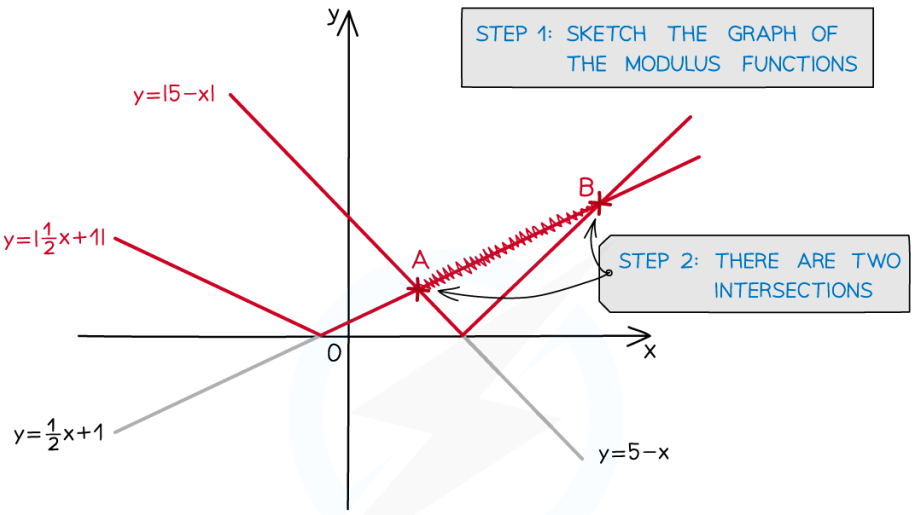
Worked Example

Your notes

(a) Solve the equation $\left|\frac{1}{2}x+1\right| = |5-x|$

(b) Hence solve the inequality $\left|\frac{1}{2}x+1\right| \geq |5-x|$

(a)



A: $\frac{1}{2}x+1=5-x$

$\frac{3}{2}x=4$

$x=\frac{8}{3}$

STEP 3: POINT A IS THE NORMAL PART OF BOTH EQUATIONS

B: $\frac{1}{2}x+1=-(5-x)$

$6=\frac{1}{2}x$

$x=12$

STEP 3: POINT B IS THE NORMAL PART OF $\frac{1}{2}x+1$ BUT THE (REFLECTED) NEGATIVE PART OF $5-x$

∴ SOLUTIONS TO $\left|\frac{1}{2}x+1\right| = |5-x|$ ARE $x=\frac{8}{3}$ AND $x=12$

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(b) $\frac{8}{3} \leqslant x \leqslant 12$

SHADE ON YOUR DIAGRAM THE PART OF THE GRAPH WHERE $|\frac{1}{2}x+1| \geqslant |5-x|$
THIS SHADING IS BETWEEN THE TWO INTERSECTIONS WHICH YOU FOUND IN PART a)

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